NIK DAVE

Senior Engineer | PhD Mechanical Engineering, ASML
Veldhoven, Netherlands

Nik.Dave@asml.com nik-dave.github.io
GitHub

PROFESSIONAL SUMMARY

Senior Engineer with a PhD in Mechanical Engineering and extensive experience in semiconductor technology, particle transport, contamination control and defectivity engineering. Skilled in combining experimental diagnostics, Particle Tracking Velocimetry (PTV), CFD, numerical modelling and data analysis to investigate complex physical processes and support technology development. Experienced in designing, executing and interpreting engineering experiments, developing analytical tools and translating technical findings into actionable recommendations for advanced manufacturing environments.

PROFESSIONAL EXPERIENCE

- **Mechanical Engineer/ Scrum Master** | ASML (via Bright Society) · *Veldhoven, Netherlands* **Jan. 2025 – Present**
  - Working within ASML Defectivity Group, supporting defectivity solutions for semiconductor manufacturing systems.
  - Owner and co-developer of the ASML Particle Tracking Tool supporting ASML designers in their contamination analysis.
  - Scrum Master of Common Defectivity Test Rig (CDTR) team. Jira setup and project management as SaFe coach of the team.
  - Collaboration with Test Measurement facility (TMF) team to perform Particle Tracking Velocimetry development.
  - Collaboration with Zeiss for MEMs defectivity study with experimental measurement and image analysis.
  - Collaboration with CFD and DUV competence for research on micro particle behaviour in atmospheric conditions in semiconductor systems.
- **Functional Design Engineer** | ASML (via Bright Society) · *Veldhoven, Netherlands* **Jan. 2023 – Dec 2024**
  - Working within ASML DUV metrology development, supporting overlay and focus drift-control solutions for semiconductor manufacturing systems.
  - Owner and developer of the BMMO NXT Focus MATLAB model supporting the IMO3 programme.
  - Supported migration activities from SVN to GitHub and acted as GitHub representative within the development team.
  - Served as Delegate Scrum Master for the Drift Control team, supporting agile delivery across functional and software engineering disciplines.
  - Organised knowledge-sharing sessions, GitHub workshops and continuous-improvement initiatives within the organisation.
- **Scientific Researcher (Doctoral Researcher)** | Eindhoven University of Technology (TU/e), Université libre de Bruxelles (ULB) & Canon Production Printing · *Eindhoven / Venlo, Netherlands & Brussels, Belgium* **Sep. 2018 – Dec. 2022**
  - Conducted doctoral research on multiscale modelling of liquid transport and dimensional stability in paper-based materials within an academic-industrial research collaboration.
  - Developed analytical, numerical and multiphysics models to investigate transport phenomena and coupled hygro-mechanical behaviour in porous materials.

- Combined physical modelling, numerical implementation and engineering analysis across multiple material and structural scales.
- Collaborated with multidisciplinary academic and industrial partners across Europe, including Canon Production Printing.
- Supported teaching activities and supervised bachelor's and master's student projects as a teaching assistant and coach.

- **Computational Engineer (Research Internship)** | **ASTUTE, Swansea University** · *Swansea, Wales, United Kingdom* **Mar. 2017 – Aug. 2017**
  - Investigated hydrodynamic behaviour of floating devices and wave-energy systems.
  - Developed an open-source simulation workflow to evaluate the performance of Wave Energy Converters (WECs).
  - Validated simulation results against commercial software tools and engineering benchmarks.
- **Senior CFD Engineer** | *Mercedes-Benz R&D (Daimler AG)* **Jan. 2015 – Aug. 2016**
  - Performed thermal management and digital-prototyping analysis for automotive systems.
  - Supported development projects involving HVAC systems, electronic cooling, heat exchangers and vehicle thermal performance.
  - Applied CFD and CAE methods using tools including Star-CCM+, Siemens NX and MATLAB.
  - Collaborated with multidisciplinary engineering teams to improve design efficiency and reduce development costs.
- **CFD & Thermal Engineer** | *Jaguar Land Rover* **Jul. 2011 – Jan. 2015**
  - Conducted underhood thermal analysis for vehicle development programmes.
  - Supported exhaust thermal management, heat protection and cooling-system optimisation activities.
  - Worked closely with aerodynamics, wind-tunnel and HVAC teams across virtual and physical test environments.
  - Developed preprocessing and modelling approaches that reduced simulation effort and improved engineering efficiency.

TEACHING EXPERIENCE

- **Mentor & Knowledge-Sharing Lead** | **ASML** · *Veldhoven, Netherlands* **2022 – Present**

  Supporting onboarding, coaching and technical knowledge transfer within engineering teams.
  - Mentored interns and engineers on GitHub workflows and development practices.
  - Organised knowledge-sharing sessions and technical workshops.
  - Supported continuous-improvement and agile team initiatives.
- **Teaching Assistant & Student Supervisor** | **Eindhoven University of Technology (TU/e)** · *Eindhoven, Netherlands* **2018 – 2022**

  Supported teaching activities and supervised bachelor and master student projects during doctoral research.
  - Coached and supervised bachelor and master student research projects.
  - Supported academic courses and practical research activities.
  - Provided guidance on engineering modelling, simulation and experimental methods.
- **Student Representative** | **Swansea University & Universitat Politècnica de Catalunya (UPC)** · *United Kingdom & Spain* **2016 – 2018**

  Elected Student Representative for the Erasmus Mundus MSc Computational Mechanics programme.
  - Represented the interests of MSc students within the programme.
  - Acted as liaison between students, lecturers and programme coordinators.

EDUCATION

- **Ph.D., Mechanical Engineering, Eindhoven University of Technology (TU/e) & Université libre de Bruxelles (ULB)** | *Eindhoven, Netherlands & Brussels, Belgium* **Sep. 2018 – Dec. 2022**
  - Joint doctoral programme within the Marie Skłodowska-Curie Innovative Training Network / European Joint Doctorate (MSCA-ITN-EJD ProTechTion), in collaboration with Canon Production Printing.
  - Doctoral research: Multiscale modelling of through-thickness liquid transport and dimensional stability of paper sheets.
- **M.Sc., Computational Mechanics, Swansea University & Universitat Politècnica de Catalunya (UPC)** | *Swansea, Wales, United Kingdom & Barcelona, Spain* **Sep. 2016 – Aug. 2018**
  - Erasmus Mundus joint master's programme completed with Distinction.
  - Awarded a fully funded European Commission Erasmus Mundus scholarship, together with Swansea University International Excellence and Go-Global scholarships.
  - Master thesis: Modelling of electromechanical transduction at finite deformation.
  - Student Representative for the MSc Computational Mechanics programme.
- **B.Eng., Mechanical Engineering, undefined** | **Jul. 2007 – Jun. 2011**
  - Graduated with Distinction and awarded a Silver Medal for academic excellence.
  - Bachelor thesis: Testing of aluminium foam castings in automotive crash management.
  - Project work included analysis of the viability of biogas as a vehicle fuel and its associated engineering challenges.

PATENTS AND PUBLICATIONS

1. **Nik Dave**, Mary Jane, and James Bond. Test 2. , 35(5):1985–2003, 2024.
2. **Nik Dave**, Mary Jane, and James Bond. Test 1. In *MoM*. American Society of Mechanical Engineers, 2021.

GRANTS

CONFERENCE LEADERSHIP & SERVICE

CONFERENCE PRESENTATIONS

Conference Papers / Oral Presentations

- **Nik Dave**, Mary Jane, and James Bond (2021). *Test 3.* .

HONORS AND AWARDS

- WCCM & ECCOMAS Congress Scholarship
- Marie Skłodowska-Curie European Commission Fully Funded Doctorate Scholarship
- Go Global Scholarship
- Erasmus Mundus Fully Funded Masters Scholarship
- Excellence Scholarship

CERTIFICATIONS

- SAFe® Scrum Master (SSM 6.0),
- Professional Scrum Master™ I (PSM I),
- SAFe for Teams,

PROFESSIONAL MEMBERSHIPS

- Member, PSM I, since 2024